# AI Applications in Research

The Case of Claude Code as a Research Workflow

Compiled summary

June 10, 2026

**Abstract**

This document summarizes the main concepts surrounding the use of artificial intelligence in empirical research, with particular emphasis on *agentic coding assistants*—and Anthropic's **Claude Code** in particular—as instruments for conducting reproducible quantitative work. It synthesizes recurring themes from Scott Cunningham's ongoing "Claude Code" series on *Scott's Mixtape Substack*, including the most recent installment, "Claude Code 54: Amnesia" (June 2026). The aim is to provide a compact conceptual map for researchers considering, or already using, AI agents in their empirical pipelines.

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# 1 Introduction: From Autocomplete to Agentic Research

Artificial intelligence in research has moved well beyond passive text generation. The current frontier is the *agentic* assistant: a system that can read and write files, execute code, run analyses, manage version control, and iterate toward a goal through extended dialogue. **Claude Code** is a prominent example—a command-line agent that operates directly inside a researcher's project directory.

For empirical disciplines such as economics, the appeal is immediate. The craft of applied work has always rested on disciplined software engineering habits that most researchers acquire slowly and imperfectly. An agentic assistant can supply those habits by default.

# 2 Core Concepts

## 2.1 The Research Workflow as Software

Modern empirical research is, in practice, a software pipeline: raw data flow through cleaning, transformation, estimation, and the automated production of tables and figures. The quality of a project is largely the quality of this pipeline. Key disciplines include:

- **Consistent naming conventions** for files and folders.
- **A well-organized, repeatable directory structure** reused across projects.
- **Automated production** of tables and figures (no manual copy-paste).
- **A pipeline of code** rather than disconnected one-off scripts.
- **Version control** (e.g., `git`) for history and reproducibility.

## 2.2 What an AI Agent Contributes

The central observation is that a capable coding agent performs the above disciplines *natively*. Claude Code creates replicable code, organizes the directory sensibly, generates figures and tables programmatically, applies consistent naming, and—because git operations are accessible to it—makes version control straightforward. For a researcher, this can lower the activation energy of good practice and, plausibly, reduce the rate of mechanical error.

## 2.3 Dialogue-Driven (Conversational) Research

A distinctive mode of use is *co-authoring the research process in real time* through conversation, rather than dictating discrete tasks or designing rigid multi-step orchestration in advance. The researcher "talks with and to" the agent. This is powerful for discovery and exploration, but—as discussed below—it shifts where and how errors are produced.

# 3 The Central Challenge: "Drift"

**Definition: Drift**

*Drift* is the subtle, often unnoticed degradation of an empirical project over time when work is mediated by an AI agent. The project's foundations move without the researcher's awareness, and the resulting errors appear *outside* the locations where an experienced researcher is trained to look.

Concrete manifestations of drift include:

- **Phantom scripts:** An analysis step is performed ad hoc during dialogue but never written down as a durable, re-runnable script.
- **Shifting analytical sample:** The sample silently changes over time, implying a script is being modified unknowingly—or that the project's foundation is coming loose.
- **Mutating treatment definitions:** Treatment units subtly differ from how they were defined weeks or months earlier.
- **Stale exhibits:** A proliferation of figures and tables, some of which have quietly become outdated, with no easy way to tell which.

### 3.1 Why Drift Is Hard to Detect

The danger is not that errors occur—errors always occur—but that they are *not caught in real time*. Two structural reasons stand out:

1. **The "well-designed sprawl" problem.** An agent's output does not look like a mess. It looks organized—"a Claude mess, not a human mess"—which makes it harder to recognize, diagnose, and repair. The familiar visual cues of disorder are absent.
2. **The erosion of memorization.** Researchers historically compensate for fallible memory through *repetition and hyper-focus*: doing the same steps repeatedly builds an internal map of the project. When the agent writes the scaffolding instead, that compensating memorization never forms. Returning to a project after even a couple of days can leave one "unusually amnesiac."

### 3.2 Drift as a New Form of an Old Problem

Drift is recognizably a descendant of pre-AI pathologies—for example, the "poor man's version control" of saving ten date-stamped copies of a file. The difference is that the AI-era version is camouflaged by tidiness, making it *harder* to notice than its messy human ancestor.

## 4 Toward Solutions: Building a "Harness"

The proposed response is not a single feature but a *system*—a personal infrastructure, or **harness**, that is the physical expression and capture of the research workflow. Several design principles emerge:

- **Simple, habitual policies over clever tooling.** The most promising fix is a small set (roughly 5–6) of policy rules that, when *habitually* followed, make errors *knowable and controllable*. A bespoke "skill" is viewed skeptically because the problem is cognitive and habitual, not merely technical.
- **Progress logs.** The agent writes a log after every major session, creating a durable trail of what changed and why.
- **The "beautiful deck."** A daily process reads the progress logs and generates a flippable HTML slide deck—with beautiful figures, tables, and slides—accessible from a dashboard side panel. Its sole purpose is to *re-orient* the researcher after time away: a deliberate antidote to amnesia.

- **External mapping (analog notebook).** A large analog notebook to map what the agent is doing, acknowledging that the agent currently has "too much latitude."

**Design Principle**

**Build systems *endogenous* to your constraints.** Rather than assuming away human limitations (memory, attention, fallibility), design the workflow so that those limitations are explicitly accommodated. Acceptance of the constraint is the precondition for a true fix rather than a band-aid.

## 5 Broader Themes for AI in Research

- **Reproducibility is necessary but not sufficient.** An agent can make code reproducible while the *conceptual* foundation (sample, treatment, identification) still drifts. Reproducibility of a moving target is not enough.
- **Agent latitude vs. control.** The more autonomy the agent has, the greater the productivity *and* the greater the surface for silent error. Effective use requires deliberate constraints on that latitude.
- **Human capital must adapt.** Researchers' error-detection instincts were trained on human-generated workflows; AI shifts the distribution of errors and requires new monitoring habits.
- **Disclosure and norms.** The series also engages with the professional dimension—how journals and the field should handle disclosure of AI use—underscoring that adoption is as much a normative question as a technical one.

## 6 Practical Recommendations

Distilling the above into actionable guidance for researchers using Claude Code or similar agents:

1. **Insist on durable scripts.** Require that every analysis step be written to a re-runnable script, never left as ephemeral dialogue.
2. **Pin the foundations.** Define and freeze the analytical sample and treatment definitions explicitly; audit them whenever the project resumes.
3. **Keep a session log.** Have the agent record what changed after each session, and review it on return.
4. **Maintain a re-orientation artifact.** Use a dashboard, deck, or notebook whose only job is to rebuild your mental model after time away.
5. **Adopt a few non-negotiable habits.** A short, fixed checklist beats elaborate tooling; consistency is what makes errors visible.
6. **Periodically rebuild from the ground up.** When drift is suspected, reconstructing a project from its foundations can be more honest than patching.

# 7 Conclusion

Agentic AI assistants like Claude Code can deliver, by default, the engineering discipline that empirical research has always demanded—reproducible code, organized projects, automated exhibits, and accessible version control. Yet the same fluency introduces a new and subtle failure mode, *drift*, in which a project's conceptual foundations move without notice precisely because the agent's output looks orderly. The path forward is not to abandon the tool but to wrap it in a deliberate *harness*: simple habitual policies, durable logs, re-orientation artifacts, and explicit constraints on agent latitude. The guiding philosophy is to build systems *endogenous* to human limitations—designing for the researcher one actually is, not the one a tool pretends them to be.

# 8 Newsletter Digest: Unread Issues, May 12 – June 10, 2026

This section summarizes the recent backlog of *Scott's Mixtape Substack* newsletters (18 distinct issues), grouped by theme. Each entry links to the web version of the post.

## 8.1 The "Claude Code" Series: AI Agents in Research

- **Claude Code 50: Claude is Holding On To Its Reasons** (May 13). Claude Code keeps a JSONL "diary" of all its reasoning and abandoned work—a goldmine for studying agent behavior. In Cunningham's minimum-wage experiments, agents randomly primed with different expected effects visibly engaged in *specification searching*: running a model, disliking the result, and silently trying another. Agents do not bind themselves to design principles or target a particular population estimand, so the human must scaffold and verify. Marks six months and fifty essays since he adopted the tool: "I am pushing all my chips onto this technology."
- **Claude Code 51: Harnessing AI Agents for Economic Research** (May 21). Atomistic `/skills` bolted onto an old workflow are not enough; the structure of research work has shifted and requires a redesigned human-in-the-loop system. Introduces the concept of an *agent harness*—the software infrastructure wrapping the model that manages context, tools, memory, and verification—and plans to rebuild his workflow from first principles, starting from David Allen's *Getting Things Done*.
- **Claude Code 53: Journal Cartels and AI Disclosure** (May 27). Academic publishing is highly concentrated (Elsevier and Wiley control roughly 61% of economics journals; an implied HHI near 2,430), so violating a publisher's AI policy could effectively bar a researcher—and their coauthors—from much of the market, echoing Becker's logic that low detection probability breeds severe punishments. His position: use AI openly, disclose it honestly, and work out disclosure duties with coauthors.
- **Claude Code 53: Applied econometrics will require a detailed checklist** (June 8). "Plan-and-let-it-rip" automation backfires: two Claude instances spent hours confidently chasing wrong conjectures when the real bug was simply a CRAN vs. GitHub version difference in the `did` package—caught only because of his deep human capital in the Callaway–Sant'Anna estimator. Advocates *structured analog workflows* (Gawande-style checklists, e.g. "Pedro's Checklist"): name the estimand in potential outcomes, tabulate cohorts, plot rollout and outcome evolutions, choose covariates methodically, and insist on **zero error**. Key claim: the returns to econometrics knowledge are *higher*, not lower, with AI, because automation is not verification.

- **Claude Code 54: Amnesia** (June 10). Summarized at length in the body of this document: introduces "drift," connects it to ADHD-related memory compensation, and describes the "beautiful deck" re-orientation artifact added to his harness.

### 8.2 AI and the Economics Profession

- **What a panel of economists said about AI in the production of research** (May 20). Writeup of an NBER panel (Simon, Cunningham, Bradford, Wing) with commentaries by Beam, Fletcher, and Goldsmith-Pinkham. Highlights: the Zurich APE project's 1,000 autonomously written papers whose right tail rates alongside published AER work; editors fear a submission flood that desk-rejection workflows cannot absorb; restricted-data contracts collide with cloud AI tools (options: synthetic data, local open-weight models, vendor agreements). Consensus: production has decoupled from verification, and incentives—not technology—will decide whether the field gets better papers or merely more.
- **Is AI Slop saliva or spit?** (June 5). Conjecture that reactions to AI-generated text mirror the "cognition of disgust": one's own chatbot conversations feel insightful (saliva) while other people's feel repugnant (spit). Proposes a randomized tournament experiment that blinds and reveals authorship of human vs. AI writings to test for "AI bias"—an experiment he has wanted to run for three years.

### 8.3 Econometrics: Difference-in-Differences Methodology

- **Should I Include Covariates in Diff-in-Diff?** (June 1). Pushes back on the belief that covariate-sensitive estimates are untrustworthy: when groups are balanced on a covariate that affects trends, unconditional parallel trends holds and no controls are needed; imbalance is what mechanically breaks parallel trends. First of a planned series (partially paywalled).
- **Diff-in-diff can be written down six ways!** (June 4). The 2x2 diff-in-diff is "four averages and three subtractions," computable two manual ways (first differences then difference; group differences then difference) and via four numerically identical OLS specifications (saturated dummies, twoway fixed effects, first-difference regression, group-difference regression)—with Stata code on Card–Krueger and castle-doctrine data to prove it.
- **Greetings from Glasgow** (May 18). A reflection on regression to the mean: the distributions of $Y(0)$ and $Y(1)$ are different variables that need not share any moments, so talk of "the outcome" reverting to "the mean" is ill-posed unless one specifies *which* potential outcome. Plus travel misadventures (a near-missed Zurich border queue and a Dublin gate change missed while talking to a chatbot).

### 8.4 The Odd Couple Podcast (with Caitlin Myers)

- **Episode 10: Doses and Decompositions!** (May 12). Walks through the decomposition of the OLS twoway fixed effects estimator for continuous-treatment difference-in-differences.
- **Episode 11: Why is our number this number?** (May 19). Continues the Frisch–Waugh–Lovell decomposition of OLS for the continuous-treatment case, working through a Claude-made deck on Callaway, Goodman-Bacon and Sant'Anna (forthcoming AER).
- **Episode 12: What's our causal effect called?** (May 26). Switches from estimation to defining causal parameters (ATT vs. ACRT, the slope of the dose-response curve) and asks

whether their abortion-access "dose" is distance in levels or in changes—an argument for starting from parameters, not regressions.

- **Episode 13: Finally the Continuous Diff-in-Diff Estimator Shows Up!** (June 9). Concludes the parameter discussion for their Texas House Bill 2 abortion-clinic study (no-anticipation pins down which dose definition is admissible) and introduces the continuous-treatment estimator itself: splines, kernels, polynomials, and when a fitted curve may be read causally.

### 8.5 Travel, Workshops, and Community

- **Weekend tabs: From Zurich** (May 16). After a three-day panel-econometrics workshop in Zurich, notes that "vibe-coded" slides eroded the memorized delivery he normally relies on—an early sighting of the human-capital theme. Link roundup spans chatbot privacy leaks, a mindfulness-app RCT, and AI-generated minimum-wage papers.
- **Day one in Madrid** (May 25). A personal travel essay: new mantras ("thoughts are not instructions"; "do the next real thing") and musings on integrity as an off-equilibrium-path concept in game theory.
- **Third Annual CodeChella Madrid is in the Books** (May 29). The diff-in-diff workshop drew its biggest crowd yet (about 150, up from 80); announces a January CodeChella in Mexico City (contact `info@mixtape.consulting`) and the 800-page book *Causal Inference: the Remix*, out August 25 with a free online version planned.
- **Closing my tabs: the Italian Job** (May 30). Travel interlude en route to Lake Maggiore; tabs include Anthropic's near-trillion-dollar valuation, the publication of the JEL "Difference-in-Differences: A Practitioner's Guide" (Baker et al.), and the Opus 4.8 release.
- **Closing tabs from Lake Maggiore in Italy** (June 6). Halfway through eight weeks in Europe after teaching at the European Commission in Ispra (Claude chose his hotel—on the wrong side of the lake, but beautifully). Tabs include Anthropic's internal productivity data, Ethan Mollick's forthcoming *Co-Existence*, NeurIPS's human-authorship policy, and a PLOS ONE RCT on bias against AI writing.

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*Source material:* Scott Cunningham, *Scott's Mixtape Substack* (`https://causalinf.substack.com`), issues of May 12 – June 10, 2026, including the "Claude Code" series on harnessing AI agents for economic research, AI disclosure norms, and reproducible empirical workflows.